KSAT Ranging Portal

Welcome to ranging!

This page introduces the radiometric products that KSAT offers on select antennas in our network. Radiometrics includes:

- range measurements
- range rate (aka Doppler)
- integrated carrier phase (aka integrated Doppler, accumulated delta range)

Support for integrated carrier phase is in development.

Supporting products include:

- angle data from the antenna controller
- signal metrics, such as signal power and signal/noise ratio
- atmospheric measurements to support troposphere models

Data products are delivered to the user in the form of CCSDS 503.0-B-2 TDM files.

Ranging operations are controlled using API calls, allowing the user to reconfigure ranging during a pass, and to collect the ranging data.

For information about the ranging API, see:

For information about TDM files, see:

- Overview: Overview of the TDM format used for KSAT ranging products.
- TRACK: Radiometric tracking data (range, carrier phase, range rate):
- ANGLE: Antenna pointing angles
- SIGMET: Signal metrics
- METEO: Meteorology

For information about how to process the data in the TDM files, see:

- ranging model
- delay model
- ranging detailed algorithm
- range rate model
